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Project:

An autonomous backend Design for a modern multi-floor parking building that replaces manual paper ticketing and random parking.

Instead of a driver wandering through basement floors looking for a spot or paying a static flat fee, This automates the entire vehicle lifecycle from entry to exit

Problem Statement:

Modern urban commercial centers, tech hubs, and malls face massive congestion and parking inefficiencies due to three main problems:

- **Space Wastage & Lack of Dimensional Intelligence:** Standard parking lots assign spots randomly or by manual ticketing. A compact hatchback often occupies an oversized SUV bay, wasting valuable clearance, while large vehicles struggle with low roof clearances or tight turns.
- **Unmanaged EV Infrastructure:** With electric vehicles (EVs) rising rapidly, drivers block charging bays without actively charging, and billing for parking time versus actual electricity drawn is rarely unified.
- **Static, Reactive Pricing:** Traditional facilities charge flat hourly rates. When the building hits peak congestion, access roads choke with traffic because pricing does not dynamically adapt to demand before gridlock occurs.

Database Design & Collections:

It consists of **4 core collections**

USERS
_id, username, email, tier, wallet_balance, [registered_fleet] (EMBEDDED)

PARKING_SLOTS
_id, floor, aisle, slot_number, size_class, dimensions features (ev_charger, disabled)

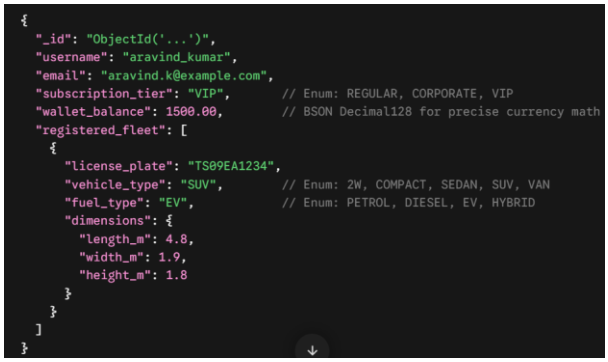
PARKING_SESSIONS
_id, user_id (Ref), slot_id (Ref), vehicle_plate, vehicle_type, lifecycle_state, ingress_time, egress_time [ev_charging_telemetry] (EMBEDDED Time-Series) {tariff_breakdown} (EMBEDDED Settlement)

TARIFF_POLICIES
_id, vehicle_category, base_rate_per_hour, ev_unit_rate_kwh,

1. users :

Stores user accounts, subscription tiers, digital wallet credits, and their owned vehicles.

Schema:

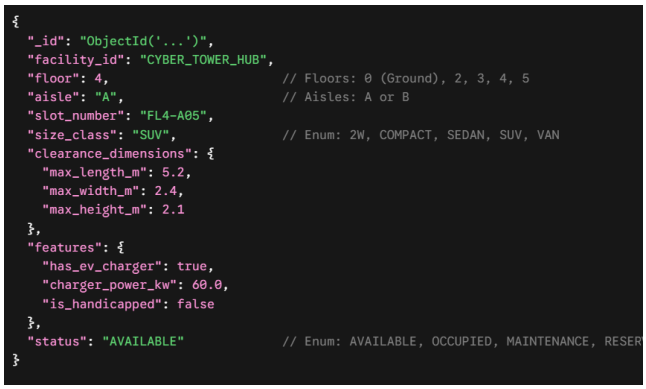


2. parking_slots :

Represents the physical facility infrastructure (100 total bays over 5 floors).

Physical bays are fixed infrastructure queried independently by status (AVAILABLE, OCCUPIED, RESERVED).

Schema:



```
{
  "_id": "ObjectId('...')",
  "facility_id": "CYBER_TOWER_HUB",
  "floor": 4,                       // Floors: 0 (Ground), 2, 3, 4, 5
  "aisle": "A",                     // Aisles: A or B
  "slot_number": "FL4-A05",
  "size_class": "SUV",              // Enum: 2W, COMPACT, SEDAN, SUV, VAN
  "clearance_dimensions": {
    "max_length_m": 5.2,
    "max_width_m": 2.4,
    "max_height_m": 2.1
  },
  "features": {
    "has_ev_charger": true,
    "charger_power_kw": 60.0,
    "is_handicapped": false
  },
  "status": "AVAILABLE"             // Enum: AVAILABLE, OCCUPIED, MAINTENANCE, RESER
}
```

3. parking_sessions :

The transactional log that records every vehicle arrival, active stay, and checkout.

Why Referenced (user_id, slot_id)? A busy facility logs tens of thousands of parking events over time. Storing sessions inside users or slots would cause unbound array growth

Why Embedded Telemetry & Tariff? Sensor readings and final line-item charges belong strictly to *that single session* and are never modified once settled.



```
{
  "_id": "ObjectId('...')",
  "user_id": "ObjectId('...')",     // Referenced User
  "slot_id": "ObjectId('...')",     // Referenced Slot
  "vehicle_plate": "TS09EA1234",
  "vehicle_type": "SUV",
  "lifecycle_state": "SETTLED",     // RESERVED -> ACTIVE -> CHECKED_OUT -> SETTLED
  "created_at": "ISODate('2026-10-28T10:00:00Z')",
  "ingress_time": "ISODate('2026-10-28T10:00:00Z')",
  "egress_time": "ISODate('2026-10-28T12:00:00Z')",
  "ev_charging_telemetry": [        // Embedded time-series array
    {"timestamp": "ISODate('...')", "power_draw_kw": 50.0, "cumulative_kwh": 0.0},
    {"timestamp": "ISODate('...')", "power_draw_kw": 25.0, "cumulative_kwh": 22.5}
  ],
  "tariff_breakdown": {             // Embedded financial settlement
    "base_rate": 80.00,
    "surge_multiplier": 1.5,
    "charging_cost": 360.00,
    "total_amount": 600.00
  }
}
```

4. tariff_policies :

Defines hourly parking rate cards, EV power costs per kilowatt-hour, and occupancy surge triggers for each vehicle category.

Schema:

```
{
  "_id": "ObjectId('...')",
  "vehicle_category": "SUV",
  "base_rate_per_hour": 80.00,
  "ev_unit_rate_kwh": 16.00,
  "occupancy_thresholds": [
    {"occupancy_pct": 70, "multiplier": 1.5},
    {"occupancy_pct": 90, "multiplier": 2.5}
  ]
}
```